

BULLETS OF FURY

Update notes for Brian

Everything below is on main at github.com/NexusWebOS/BulletsOfFury. Written so you can pick the repo up without having to reverse-engineer what moved.

THE BIG ONE — every stage was being blurred

`ctx.imageSmoothingEnabled` was being set to `TRUE` on the per-frame render setup. With `SS=2` that bilinear-doubled every 800-wide master, and the canvas CSS was `image-rendering:auto`, so the browser blurred it a second time on the way to the display.

Measured on stage 1: one backdrop crop went from 21,423 distinct colours to 13,927, and 192/192 sampled 2px column pairs are now identical — a true integer double. If you touch the render path, do not re-enable smoothing.

GAMEPLAY FIXES

- Flame / ice hitboxes now match the drawn plume. They were 96px wide on screen and 35px in the hit test at level 5, so point-blank fire did nothing.
- Ice breath was anchored at the wrong end — it floated 27px clear of the ship. The tip pulls back now, not the nozzle.
- Roadside signs and stage props were travelling UP while the terrain travelled DOWN. Both now read a shared `_masterSrcY`.
- Tanks were being shoved sideways by `enemyEntrySweep`, which runs OUTSIDE their pattern. Gated on `_vkind`. 62 lateral writes -> 0.
- Level 7: 1,577 purple halo pixels converted to a black edge (alpha preserved).
- Stage 2 boss: 198 white edge pixels converted to black.
- Missile-lock beep removed; flame/ice now make a hit sound.

Stages & bosses

BACKGROUNDS

- Stage 5 was running storm800_rc2_master — a STORM plate, cloud and rain for nearly all 5120px. It is a space stage. Now loops norb5_arena.
- Stage 6 is back on nsky6_sky, looped. The RC2 rebuild had reintroduced a fortress door and platform decks that Mike had already ruled out.
- Stage 6 weather: the darkening was never actually written — l6WeatherDraw drew rain and lightning only. It looked right purely because the old master had a night sky painted in. Now dims 177.6 -> 134.5 luminance from clear to storm.
- Stage 8 skips master rows 2500-2650 to hide a bridge that is PAINTED INTO the plate.
- Stage 4: roadside signs and the car pileup removed.

MINIBOSSES — all eight slots filled

STAGE	MINIBOSS	NOTE
1	junglecruiser	
2	MAGMA WARD	new
3	RIME WALL	new
4	OLIVE WARDEN	new
5	subcore	
6	blacksteel	moved from 4
7	ratking	see below
8	herald	

STAGE 7 IS DELIBERATELY NOT SWAPPED. Mike asked for olivecarrier there. Putting it in stopped the stage-7 sub-boss triggering at all — the soak observation _sbSeen went false and the boss gate behind it failed with it. ratking is a SEWER unit and stage 7 appears to depend on that. There is an assertion pinning this with the reason; please do not "fix" it back.

Known broken / in flight

DO NOT ASSUME THESE ARE DONE

- Vol.2 bosses (Xeno Regent / Doomsday Carrier / Sludge Emperor) are imported and BUILD correctly, but render as a dark polygon instead of their art. Stages 5/6/7 are reverted to their old working bosses so nothing ships broken. Flip the three STAGES entries once the draw is understood.
- Space/sky background props (satellites, debris) have OPAQUE cell backgrounds — they show as a rotated box around each object. Needs an alpha key on those cells.
- The old chain-lightning graphic still appears in stage 8 scrolling terrain.
- Stages 1, 2 and 7 do not match their own subtitles — stage 1 turns to desert after one screen, stage 2 ("IT'S HOT IN HERE") is nearly all pale sand, stage 7 has a large black void at scroll 4000.
- Craters on stage 4, the stage-6 door and the stage-8 bridge are PAINTED INTO the masters. They cannot be removed in code — they need art regeneration or a sampling patch.
- The bottom hint bar was restyled but is NOT visually verified.

FIVE SUITE FAILURES ARE LONG-STANDING

preload count, two _superseded ledger checks, volley round count, flash families. A clean run is 2,647 assertions / 234 sections / 5 failures. If you see MORE than five, check git status for deleted art before debugging anything — a missing file trips nine assertions across four sections and reads as an unrelated pile of failures.

Traps that have each cost a session

These are not hypothetical. Every one produced a confident wrong answer first.

Filenames lie. Render before you trust.

slatejet is debris, not jets. nsw_dist is a distortion effect. The nhb_boss "ICE" and "LAVA" entries are 368x78 HEALTH BAR rigs, not units. storm800_rc2_master is a storm plate on a space stage. Resolve a MID-REEL frame and look at it.

XART.rdy(key) is FALSE on the first call — it STARTS the decode.

A single synchronous render draws nothing and looks exactly like missing art. Warm the key, then wait, then draw.

There are TWO boss spawners with near-identical switches.

spawnSubBoss__inner is minibosses; spawnBoss is stage bosses. Registering a kind in the wrong one gives you a 160x120 unit with no name and placeholder art. The tell: spawnBoss on a unit that works in game fails the same way.

BOFFI rects are [sheetIndex, x, y, w, h] — FIVE elements.

Reading them as four resolves every rect to None. Atlas sheets are SHARED, so a wrong rect corrupts art far outside what you were editing.

A green suite proves state, not pixels.

_BUILD_SOURCE/shoot.py and the probe_*.py scripts drive real Chromium and read pixels off the canvas. Several bugs above were green in the suite the whole time.

Check that your probe measured SOMETHING.

A probe here once printed "signs ride with the terrain" having measured zero stages. Another reported eight boss candidates at identical dimensions — that was the generic fallback, not a roster.